OFFICIAL USER GUIDE

# EconWebCast

## AI-Powered Web-Based Data Analysis and Prediction Platform

*A research analytics application for econometric modeling and forecasting*

Application link: https://econ-web-cast.vercel.app/

| Field | Detail |
| --- | --- |
| Application name | EconWebCast |
| Document type | Official Application User Guide |
| Audience | Researchers, economists, and data analysts |
| Access | Web browser (no installation required) — econ-web-cast.vercel.app |
| Sign-in | Google Account |
| Version | 1.0 |
| Contact | kbretmanna@gmail.com |

### A note before you begin

EconWebCast is an actively developed application, and new features, models, and refinements are added over time. Your feedback directly shapes future updates. If you have questions, run into an issue, or have suggestions for improvement, please email kbretmanna@gmail.com — comments from researchers using the platform are highly encouraged and genuinely welcomed.

## 1. Application Overview

**EconWebCast** (available at https://econ-web-cast.vercel.app/) is the AI-Powered Web-Based Data Analysis and Prediction platform — an online research tool that lets you run professional statistical and machine-learning models on your own datasets, without writing any code.

You upload a spreadsheet (a CSV file), point and click to tell the application which columns are your outcome and your predictors, choose a statistical method, and the platform estimates the model on secure servers and returns your results in clear tables and charts. Results can be downloaded as a ready-to-present PowerPoint deck.

The application is organized around the three data structures most common in economic and social-science research:

| Module | Use it when your data is... |
| --- | --- |
| Cross-Sectional Data | Many subjects observed once (e.g. 500 firms surveyed this year) |
| Time Series Data | One subject observed repeatedly over time (e.g. monthly inflation for 10 years) |
| Panel Data | Many subjects observed over multiple time periods (e.g. 50 countries over 20 years) |

**Note**

This is a research and estimation tool. It supports your analysis and reporting; it does not replace your judgment as a researcher about which model is appropriate for your question.

## 2. Purpose and Scope

### Purpose

EconWebCast exists to make rigorous econometric estimation accessible, fast, and reproducible for researchers who understand statistics but may not program. It removes the software-engineering barrier — no Python, R, or Stata installation, no package management, no scripting — while still providing the diagnostic depth that credible empirical work requires.

### In scope

- Uploading tabular research data (CSV format).
- Descriptive/summary statistics for selected variables.
- Estimating regression and machine-learning models across cross-sectional, time-series, and panel data.
- Automatic model diagnostics (tests for heteroskedasticity, multicollinearity, autocorrelation, normality, model specification).
- Forecasting and prediction with out-of-sample validation.
- Visualizing a time series as a line graph.
- Exporting formatted results to PowerPoint.

### Out of scope

- EconWebCast does not collect data for you or clean it beyond the operations in the Data Cleanup module.
- It does not choose the "correct" model for your research question — you select the method.
- It is not a general-purpose spreadsheet, database, or data-warehousing tool.
- It does not currently accept file formats other than CSV.

## 3. Intended Users and Use Cases

### Who this guide is for

This guide assumes you are a researcher, economist, extension specialist, graduate student, or analyst who:

- Understands core statistical ideas (regression, significance, $R^2$, etc.).
- Works with structured, tabular data.
- Wants credible results quickly, without programming.

No programming experience is assumed or required.

### Representative use cases

| User | Typical goal |
|---|---|
| Applied economist | Estimate the effect of education on wages using survey data (cross-sectional OLS). |
| Macro/finance researcher | Forecast a price or index over the next periods (time series ARIMA). |
| Development/policy researcher | Measure a policy's impact across countries over time (panel fixed effects). |
| Graduate student | Produce diagnostic-checked regression tables and slides for a thesis chapter. |
| Data analyst | Compare several models (OLS vs. LASSO vs. Random Forest) on the same dataset. |

## 4. Key Concepts and Terminology

To keep the rest of this guide clear, here are the terms used consistently throughout, in plain language.

| Term | Plain-language meaning |
|---|---|
| Dependent variable | The outcome you are trying to explain or predict (e.g. wage, price, GDP). Also called the endogenous or target variable. |
| Independent variable | A predictor you believe influences the outcome (e.g. years of schooling). Also called a feature or regressor. |

| Term | Plain-language meaning |
|---|---|
| Exogenous variable | In time series, an outside predictor used alongside time (e.g. temperature when forecasting energy demand). |
| ID column | The column that uniquely labels each subject (person, firm, country). Used to identify rows, not as a predictor. |
| Categorical variable | A predictor made of categories rather than numbers (e.g. region, industry). The app converts these to numbers automatically. |
| Coefficient | A number showing how much the outcome changes when a predictor increases by one unit. |
| p-value | The probability the observed relationship is due to chance. Small (below 0.05) usually means "statistically significant." |
| $R^2$ (R-squared) | The share of variation in the outcome explained by the model, from 0 to 1. Higher generally means a better fit. |
| Feature importance | For tree-based models, a ranking of which predictors mattered most (used in place of coefficients). |
| Diagnostic test | An automatic check on whether the model's statistical assumptions hold. |
| Cross-validation (CV) | Repeatedly testing the model on data it did not train on, to gauge how well it generalizes. |

## 5. System Workflow Overview

Every analysis in EconWebCast follows the same five-stage flow, regardless of module:

| Stage | Description |
|---|---|
| 1. Sign In | Sign in with your Google Account. |
| 2. Choose a Module | Choose a module from the dashboard that matches your data structure. |
| 3. Upload & Configure | Upload your CSV and configure the analysis (select method and variables). |
| 4. Run Model (Predict) | Run summary statistics, a line graph, and/or the model (Predict). |
| 5. Review & Export | Review the tables and diagnostics, then export to PowerPoint. |

**Tip**

Every analysis screen has a "Guide Me" button in the top-right corner that opens step-by-step help specific to that screen. This user guide expands on that in-app help.

## 6. Getting Started: Signing In

EconWebCast requires a Google sign-in. This protects your session and your data.

### Procedure — Signing in

1. Open the application in your web browser at econ-web-cast.vercel.app.
2. On the Welcome screen, locate the "Sign in with your Google Account" button.
3. Click the Google sign-in button.
4. In the Google window, choose the account you want to use and approve access.
5. On success, you are taken automatically to the Dashboard.

What you should expect: Your first name and email appear in a small user card in the top-right of the dashboard, confirming you are signed in.

> **Note**
> If your session expires while you work, the application may ask you to sign in again. This is normal security behavior — simply sign back in and continue.

### Signing out

Click the red Logout button in the user card (top-right). You are returned to the Welcome screen.

## 7. The Dashboard

After signing in you arrive at the Dashboard, titled "AI-Powered Web-Based Data Analysis and Prediction."

### What you see

- A heading and the prompt "Select a data category to continue."
- Three large clickable cards: Cross-Sectional Data, Time Series Data, Panel Data.
- A "Guide Me" button (top-right) with dashboard-level help.
- Your user card with your name, email, and Logout.

### Procedure — Opening a module

6. Decide which data structure matches your dataset (see the table in Section 1).
7. Click the matching card.
8. The module's analysis screen opens, ready for you to upload a file.

To return to the dashboard from any module, click the back arrow (←) in the top-left of the module's header.

## 8. Preparing Your Data

Good input data is the single biggest factor in getting trustworthy results. Please read this section before your first upload.

### File format requirements

| Requirement | Detail |
|---|---|
| File type | CSV (comma-separated values). Export from Excel/Sheets using Save As → CSV. |
| First row | Must contain column names (headers). These names populate the app's dropdown menus. |
| One observation per row | Each row is a single record (a person, firm, country, or a single time point). |
| Consistent columns | Every row should have the same set of columns. |

### Content guidelines

- Use clear, unique column names (e.g. wage, education_years, region) — you will select them by name in dropdowns.
- Numbers should be numeric, not text. Avoid currency symbols, thousands separators, or stray spaces inside numeric columns (e.g. use 1250, not $1,250).
- Dates (for time series and panel data) should be in a recognizable format such as 2023-01-31, 01/31/2023, or Jan 2023. The app validates that the column can be read as dates.
- Missing values are allowed, but each module handles them.

### Data-size guidance

- Time-series estimation requires enough history to validate the model. As a rule of thumb, provide at least 10 observations, and more when you increase the number of cross-validation folds.
- Very large files take longer to upload and estimate. Start with a focused dataset containing only the columns you need.

> **Tip**
>
> Before running a model, always click "Summary Statistics" first. It's a fast sanity check that your columns were read correctly (reasonable means, min/max, and counts).

## 9. How EconWebCast Adapts Models Across Data Structures

LASSO, Ridge, Bagging, Random Forest, and Gradient Boosting are available in all three modules below (Cross-Sectional, Time Series, and Panel), and several diagnostic and validation ideas carry across modules as well. Before the module-by-module instructions, this section explains what changes — and why — when the same statistical idea is applied to a different data structure, so the more detailed, method-by-method sections that follow (Sections 10.4, 11.4, and 12.4) do not need to repeat this reasoning for every method.

### 9.1 Why the Same Model Behaves Differently Across Data Structures

Two data-structure-specific concerns drive nearly every implementation difference documented in this section. First, feature scaling: the LASSO and Ridge penalties are scale-dependent, so predictors are standardized (mean zero, unit variance) before fitting in the Cross-Sectional and Time Series modules (Hastie, Tibshirani, & Friedman, 2009); tree-based methods (Bagging, Random Forest, Boosting) are scale-invariant by construction and are never scaled. Second, validation design: cross-validation must respect whatever dependence structure the data has, or performance estimates become optimistic and misleading (Kohavi, 1995). Cross-sectional rows are assumed independent, so ordinary shuffled k-fold cross-validation is appropriate. Time series rows are temporally dependent, so walk-forward, non-shuffled splitting is required (Bergmeir & Benítez, 2012). Panel rows are both temporally and cross-sectionally dependent (repeated observations of the same entity over time), which requires either a fixed-effects transformation for linear models or an entity-aware validation split for tree-based models (Wooldridge, 2010).

## 9.2 Validation Design by Data Structure

| Data structure | Feature construction | Cross-validation scheme | Reportable model |
|---|---|---|---|
| Time Series | Three lags of the target (+ optional exogenous variables) | Walk-forward (TimeSeriesSplit) — trains only on the past | Refit on a chronological 80/20 split (most recent 20% held out) |
| Cross-Sectional | Independent/categorical variables as configured, no lags | Shuffled k-fold (or stratified k-fold for binary outcomes) | Refit on a random 80/20 split, stratified if binary |
| Panel — LASSO/Ridge | Within (fixed-effects) transformation: each entity's group mean subtracted from y and X | Time-based walk-forward when a usable date column exists; entity-based GroupKFold otherwise, using each fold's training rows only to compute demeaning means | Refit on the full within-transformed sample; entity fixed effects recovered separately |
| Panel — Bagging/Forest/Boosting | Entity ID one-hot encoded as additional features; level (undemeaned) data retained | Time-based walk-forward or entity-based GroupKFold — both treated as equally valid, since trees do not require demeaning to avoid omitted-variable bias | Refit on the full sample with entity dummies; entity-dummy importances collapsed into a single reported figure |

Two panel-specific design points are worth stating explicitly. First, linear models (LASSO, Ridge) require the within transformation because, without it, a fixed but unobserved characteristic of each entity would be absorbed into the error term and bias the coefficient estimates — the classical omitted-variable problem that fixed-effects estimation exists to solve (Wooldridge, 2010; Baltagi, 2021). Second, tree-based ensembles do not share this bias mechanism, so demeaning is unnecessary for them and would discard usable level information (Athey & Imbens, 2019). Consequently, entity-based cross-validation is a genuinely weaker, average-fixed-effect approximation for LASSO/Ridge (since an unseen entity's individual fixed effect cannot be estimated from zero observations of it), but is an equally valid strategy for Bagging, Forest, and Boosting, which can produce a reasonable prediction for an unseen entity from its other feature values alone.

# 10. Cross-Sectional Data Analysis

Use this module when you have many subjects observed at a single point in time — for example, a survey of 1,000 households this year.

## 10.1 Available Methods

| Method (menu label) | Best used for |
|---|---|
| OLS | Standard linear regression for a continuous outcome. The default starting point. |
| General Least Square (GLS) | Linear regression when errors are unequal or correlated. |
| Logit | A binary outcome (yes/no, 1/0), e.g. employed vs. not. |
| LASSO | Regression with many predictors; automatically shrinks weak ones toward zero. |
| RIDGE | Regression with many correlated predictors; stabilizes the estimates. |
| FOREST | Random Forest — a flexible tree-based model for non-linear relationships. |
| BOOSTING | Gradient Boosting — a powerful tree-based model built in stages. |
| BAGGING | An ensemble of trees averaged to reduce variance. |

This table summarizes when to use each method. For the underlying methodology, constants, and citations behind every method, see **Section 10.4** below.

## 10.2 Screen Elements

| Element | What it does |
|---|---|
| Upload File | Opens a file picker; select your .csv. Column names load into the dropdowns. |
| Methods | Choose the estimation method (table above). |
| ID Column | The unique row identifier (not used as a predictor). |
| Dependent Variable | The outcome you want to explain or predict. |
| Independent Variables | One or more predictors (multi-select). |
| Categorical Variables | Any predictors that are categories; encoded automatically (multi-select). |
| Outliers | Choose Yes to detect and treat extreme values before estimating, or No to leave data as-is. |
| Summary Statistics (button) | Shows descriptive stats for your selected variables. |
| Predict (button) | Runs the model. Becomes active only when required fields are set. |

| Element | What it does |
| --- | --- |
| Download as PowerPoint | Appears after a successful run; exports a formatted deck. |

## 10.3 Procedure — Running a Cross-Sectional Model

9. Click a Cross-Sectional Data card on the dashboard.
10. Click Upload File and select your CSV. Wait for the dropdowns to populate.
11. In Methods, choose your estimation method (e.g. OLS).
12. Set the ID Column to your unique identifier.
13. Set the Dependent Variable to your outcome.
14. Add one or more Independent Variables.
15. If any predictors are categories (e.g. region), add them under Categorical Variables.
16. Choose an Outliers option (Yes or No).
17. (Recommended) Click Summary Statistics and confirm the numbers look sensible.
18. Click Predict. A loading indicator appears while the model runs.
19. Review the results table and diagnostics (Section 10.5).
20. (Optional) Click Download as PowerPoint.

> **Note**
> The Predict button stays greyed-out until an ID column, dependent variable, at least one independent variable, and an outlier choice are all selected. This prevents incomplete runs.

## 10.4 Estimation Methods: Constants and References

This section documents, for every method listed in Section 10.1, the specific constants (hyperparameters), validation design, diagnostic tests, and methodological literature the implementation draws on — in the order each method appears in the Methods menu.

### *10.4.1 Ordinary Least Squares (OLS)*

OLS is EconWebCast's baseline cross-sectional regression model, fit via statsmodels and accompanied by the platform's most extensive diagnostic suite. Both in-sample fit statistics (R-squared, adjusted R-squared, F-statistic, AIC, BIC) and genuine out-of-sample performance on a held-out 20 percent test split are reported, together with a supplementary 5-fold cross-validation robustness check that re-partitions the full dataset independently of the single 80/20 split (Kohavi, 1995). Because plain OLS has no regularization strength to tune, this cross-validation exists purely to show whether the single-split test metrics are a stable estimate, not to search for a hyperparameter.

| Constant | Value | Rationale |
| --- | --- | --- |
| Train/test split | Random 80/20, random_state = 42 | Cross-sectional rows are independent; no temporal order to preserve. |

| Constant | Value | Rationale |
|---|---|---|
| Underdetermination guard | Training rows must exceed the number of parameters (independent + categorical dummy variables + intercept) | OLS with as many or more parameters than observations produces a degenerate, unreliable fit. |
| Robust standard errors | HC3 (heteroskedasticity-consistent) | Recommended general-purpose choice for cross-sectional regression, particularly in smaller samples (Long & Ervin, 2000). |
| Supplementary cross-validation | Shuffled 5-fold, on the full dataset | Stability check on the single-split test metrics, not hyperparameter selection (Kohavi, 1995). |

OLS is accompanied by seven automated diagnostic tests: heteroskedasticity (Breusch-Pagan and White tests) (Breusch & Pagan, 1979; White, 1980); multicollinearity (Variance Inflation Factor and pairwise correlation) (O'Brien, 2007); normality of residuals (Jarque-Bera and Shapiro-Wilk) (Bera & Jarque, 1980; Shapiro & Wilk, 1965); autocorrelation (Durbin-Watson) (Durbin & Watson, 1950); model specification (RESET) (Ramsey, 1969); influential observations (Cook's Distance) (Cook, 1977); and a fitted-value-versus-residual linearity check. Researchers should note that the Durbin-Watson statistic presumes a meaningful row ordering and is most informative for time-ordered data; on genuinely cross-sectional data (independent subjects with no natural order) a significant result is less interpretable and should be read with caution. If the heteroskedasticity tests flag a violation, the robust_standard_errors_hc3 and robust_p_values_hc3 fields — not the ordinary standard errors — are the ones that should be reported.

### 10.4.2 Generalized Least Squares (GLS)

EconWebCast's GLS module implements Feasible GLS (FGLS) for heteroskedastic errors, computed via weighted least squares (WLS) — weights equal to 1/variance are mathematically equivalent to GLS with a diagonal error covariance matrix, without ever constructing an N-by-N matrix (Greene, 2018, Ch. 9). Specifically, EconWebCast follows Harvey's (1976) multiplicative heteroskedasticity approach: an initial OLS fit generates residuals, the log of squared residuals is modeled as a function of the same predictors X, and the exponentiated fitted values become the WLS weights (Harvey, 1976). This three-step procedure is re-estimated independently within every cross-validation fold and on the final train/test split, so no information about held-out rows leaks into how training rows are weighted.

| Constant | Value | Rationale |
|---|---|---|
| Variance model | log(residual^2) regressed on X (multiplicative heteroskedasticity) | The log transform keeps the auxiliary regression well-behaved and guarantees a strictly positive variance estimate once exponentiated (Harvey, 1976). |
| Numerical floor | min_variance = 1e-6 | Prevents log(0) and division-by-zero in edge cases; a technical safeguard, not a substantive modeling choice. |
| Cross-validation | Shuffled k-fold, default 5 folds | Same robustness-check rationale as OLS (Kohavi, 1995). |
| Diagnostics | Not currently run for GLS | Unlike OLS, no heteroskedasticity/multicollinearity/normality/specification tests are reported alongside GLS output; researchers |

| Constant | Value | Rationale |
|---|---|---|
| | | wanting the full diagnostic picture should also run OLS on the same data for comparison. |

Because OLS is unbiased but inefficient under heteroskedasticity (Wooldridge, 2019), GLS will generally resemble OLS closely when the Breusch-Pagan/White tests (available in the OLS module on the same data) do not detect significant heteroskedasticity, and will diverge from OLS — with generally tighter standard errors — when they do.

### *10.4.3 Logit*

Logit models a binary (0/1) outcome via maximum likelihood, fit directly through statsmodels' Logit class rather than the mathematically equivalent GLM(Binomial, logit link) formulation, to match the notation and output researchers and reviewers expect (Hosmer, Lemeshow, & Sturdivant, 2013). Results are reported on three different scales — log-odds coefficients, odds ratios (exponentiated coefficients), and marginal effects at the mean — which answer different questions and should not be conflated.

| Constant | Value | Rationale |
|---|---|---|
| Train/test split | Stratified random 80/20, random_state = 42 | Stratification preserves class balance in both sets, important for imbalanced binary outcomes. |
| Marginal effects | get_margeff(at="mean", dummy=True) | dummy=True is essential: for 0/1 predictors, the correct marginal effect is the discrete change P(y=1\|d=1) - P(y=1\|d=0), not a derivative, which is only valid for continuous regressors (Wooldridge, 2019). |
| Robust standard errors | HC0 (White/sandwich estimator), requested at fit time | HC3 is a linear-model-specific small-sample correction and is not a standard robust covariance type for MLE-based discrete models; HC0 is the conventional robust SE for logit (White, 1980). |
| Convergence check | Verified defensively across multiple statsmodels attribute locations; reported as unknown (not assumed true) if unconfirmed | Avoids silently treating an unverified fit as converged. |

Diagnostics include VIF/correlation-based multicollinearity checks (O'Brien, 2007); normality tests on Pearson residuals, explicitly labeled as reference-only since normally distributed residuals are not a Logit assumption (McCullagh & Nelder, 1989); Cook's Distance for influential observations (Cook, 1977); the Hosmer-Lemeshow goodness-of-fit test, which bins predicted probabilities and compares observed to expected outcome counts within each bin (Hosmer & Lemeshow, 1980); and a separation-risk check that flags coefficients with implausibly extreme magnitude or standard error, a common symptom of (quasi-)complete separation with sparse dummy predictors (Albert & Anderson, 1984). Where separation is flagged, Firth's penalized logistic regression is the standard bias-reduction remedy (Firth, 1993; Heinze & Schemper, 2002). Model fit is additionally summarized via deviance, Pearson chi-squared, and McFadden's pseudo-R-squared (McFadden, 1974), none of which are on the same scale as OLS's R-squared and should not be compared to it directly.

### *10.4.4 LASSO Regression*

LASSO (Least Absolute Shrinkage and Selection Operator) adds an L1 penalty to ordinary least squares, which shrinks weak coefficients and can set some exactly to zero, performing variable selection and estimation simultaneously (Tibshirani, 1996). EconWebCast fits LASSO via coordinate descent (Friedman, Hastie, & Tibshirani, 2010).

The cross-sectional module auto-detects whether the outcome is continuous or binary (0/1) and fits a different model accordingly.

| Constant | Value | Rationale |
|---|---|---|
| Alpha grid | 100 candidate values (continuous branch) | Full-resolution cross-validated selection of regularization strength. |
| Solver (binary branch) | saga | The solver required for L1-penalized logistic regression at scale (Defazio, Bach, & Lacoste-Julien, 2014). |
| C grid (binary branch) | 20 log-spaced values; C = 1/alpha | Selected by maximizing held-out log-likelihood, the standard criterion for probabilistic classifiers. |
| Cross-validation | Shuffled 5-fold (StratifiedKFold for binary outcomes) | Cross-sectional rows are independent; stratification preserves class balance (Hosmer, Lemeshow, & Sturdivant, 2013). |
| Reported binary metrics | Accuracy, ROC-AUC, log loss, Brier score | Brier score evaluates probability calibration, not just classification (Brier, 1950). |

### *10.4.5 Ridge Regression*

Ridge regression adds an L2 penalty to ordinary least squares, shrinking all coefficients toward zero without setting any exactly to zero (Hoerl & Kennard, 1970). Unlike LASSO, Ridge retains every predictor in the model, which makes it a preferred choice when predictors are highly correlated (Hastie et al., 2009).

In the current release, the cross-sectional continuous-outcome branch fits Ridge at a fixed alpha = 1.0 rather than selecting it by cross-validation, and does not standardize predictors before fitting. Researchers should treat coefficient magnitudes from this branch with caution when predictors are on very different numeric scales, and should note that the regularization strength was not tuned to the specific dataset (Hoerl & Kennard, 1970). The binary-outcome branch fits an L2-penalized logistic regression (solver: lbfgs, C = 1.0, also fixed rather than cross-validated) on standardized predictors, and reports odds ratios (exp of each coefficient) alongside accuracy, ROC-AUC, log loss, and Brier score (Hosmer et al., 2013).

### *10.4.6 Bagging (Bootstrap Aggregating)*

Bagging fits many decision trees, each on an independent bootstrap sample of the training rows, and averages their predictions. Averaging independent, high-variance trees reduces prediction variance without materially increasing bias (Breiman, 1996). Tree diversity — and therefore the benefit of averaging — depends on the base trees seeing different subsets of rows and, ideally, columns; a bagging ensemble whose trees each see essentially the full dataset provides little variance reduction over a single tree.

The cross-sectional Bagging module reports two independent validation figures: an out-of-bag (OOB) estimate, computed for free from the roughly 36.8 percent of rows each tree does not see during its bootstrap sample (Breiman, 1996, 2001), and a supplementary external k-fold cross-validation using a lighter ensemble (50 trees per fold rather than the full 100) to control runtime. The two can disagree because OOB estimates performance from bootstrap resampling of a single training set, while k-fold cross-validation re-partitions the full dataset; a large disagreement between the two is itself a useful diagnostic of instability. Feature importances are computed by averaging each base tree's native impurity-based importance, together with their standard deviation across trees, which gives a sense of how consistently each predictor matters across the ensemble.

### *10.4.7 Random Forest*

Random Forest extends Bagging by adding a second decorrelation mechanism: at every individual split (not just once per tree), only a random subset of candidate predictors is considered (Breiman, 2001). This per-split feature randomization is what distinguishes Random Forest from Bagging and is generally the more effective decorrelation approach (Hastie et al., 2009, Ch. 15).

This module reports an out-of-bag $R^2$/accuracy estimate alongside a supplementary external k-fold cross-validation, following the same two-layer design as cross-sectional Bagging, with the binary-outcome branch additionally reporting importance-variability across trees.

### *10.4.8 Gradient Boosting*

Gradient Boosting builds trees sequentially, with each new tree fit to the residual errors of the ensemble built so far, rather than averaging independent trees built in parallel (Friedman, 2001). Because errors compound additively across many sequential trees, each individual tree is intentionally kept shallow — a weak learner — and the ensemble's overall strength comes from the additive, error-correcting sequence rather than from averaging.

The cross-sectional module uses a faster learning rate (0.1, the sklearn default) and a lower estimator cap (200) than the Time Series module's more conservative settings (0.05, 300). Because subsample < 1.0 is used, the model also exposes oob_improvement_, a free, per-iteration diagnostic of loss reduction on out-of-bag rows (Friedman, 2002). This is a loss-improvement trend, not a directly interpretable $R^2$ or accuracy figure — the supplementary external k-fold cross-validation section should be used for a directly interpretable estimate of out-of-sample performance.

## 10.5 Reading Your Results

After you click Predict, the results view varies by the type of method you chose.

### *Regression-based methods (OLS, GLS, Logit, LASSO, Ridge)*

You will typically see:

| Output | How to read it |
|---|---|
| Coefficient | The estimated effect of each predictor on the outcome. Sign matters: positive = raises the outcome, negative = lowers it. |
| Standard error | The uncertainty around each coefficient. Smaller is more precise. |
| p-value | Below 0.05 is commonly treated as statistically significant. |
| Significance label | The app labels each result "significant" or "not significant" at the 0.05 threshold. |

| Output | How to read it |
| --- | --- |
| Fit statistics (e.g. $R^2$) | How much of the outcome's variation the model explains. |

#### *Tree-based methods (Random Forest, Gradient Boosting, Bagging)*

Tree models don't produce coefficients. Instead you'll see:

- Feature importance — a ranking of which predictors contributed most to the predictions.
- Performance metrics — such as $R^2$, RMSE (root mean squared error), and MAE (mean absolute error). For RMSE and MAE, smaller is better.

Diagnostic tests run by each method — and what a significant result means — are described alongside that method in Section 10.4. Every test ends in a plain-language conclusion (for example, "heteroskedasticity detected" or "no significant heteroskedasticity detected"), so you do not need to interpret raw test statistics yourself.

## 11. Time Series Analysis

Use this module when you have repeated observations over time — for example, monthly sales or daily prices.

### 11.1 Available Methods

| Method | Best used for |
| --- | --- |
| ARIMA | The classic forecasting model for trend and seasonality over time. |
| LASSO / RIDGE | Regularized regression using time and optional outside predictors. |
| FOREST / BOOSTING / BAGGING | Tree-based ensemble models adapted for time-ordered data. |

This table summarizes when to use each method. For the underlying methodology, constants, and citations behind every method, see **Section 11.4** below.

### 11.2 Screen Elements

| Element | What it does |
| --- | --- |
| Upload File | Select your .csv; column names load into the dropdowns. |
| Methods | Choose the forecasting method (table above). |
| Date Variable | The column holding dates/timestamps. The app checks it contains valid dates. |
| Start Date / End Date | Populate automatically from your data; narrow the range to model a specific period. |
| Endogenous Variable | The target you want to forecast. |
| Exogenous Variables | Optional additional predictors (multi-select). |
| Summary Statistics (button) | Descriptive stats for the selected variables. |

| Element | What it does |
|---|---|
| Line Graph (button) | Plots your target variable over time. |
| Predict (button) | Runs the forecast; active only when required fields are set. |
| Clear (button) | Resets the whole screen for a new dataset. |
| Download | Exports the forecast to PowerPoint after a run. |

## 11.3 Procedure — Producing a Forecast

21. Open the Time Series Data module.
22. Upload your CSV.
23. Choose a Method (e.g. ARIMA).
24. Select the Date Variable. If the column isn't valid dates, a red warning appears and you cannot proceed. Pick a different column or fix the data.
25. Confirm or adjust the Start Date and End Date (they default to your full range).
26. Select the Endogenous Variable (your forecast target).
27. (Optional) Add Exogenous Variables.
28. (Recommended) Click Line Graph to visually inspect the series for trends, gaps, or spikes, and Summary Statistics for a numeric check.
29. Click Predict.
30. Review the forecast and diagnostics.
31. (Optional) Download the PowerPoint.
32. Click Clear to start over with a new file.

> **Tip**
> The line graph is the fastest way to catch a data problem — a flat line, a sudden jump, or a gap usually signals a formatting or missing-value issue worth fixing before forecasting.

## 11.4 Estimation Methods: Constants and References

This section documents, for every method listed in Section 11.1, the specific constants (hyperparameters), validation design, and methodological literature the implementation draws on.

### *11.4.1 ARIMA*

EconWebCast's ARIMA module fits a SARIMAX model with a fixed order of (1, 0, 1) — one autoregressive term, no differencing, one moving-average term — for every dataset submitted; order is not searched or selected by an information criterion, and applying exogenous variables produces an ARIMAX label automatically rather than through a statistical test of their usefulness. An Augmented Dickey-Fuller test is run and its result (stationary/non-stationary, p-value) is returned (Dickey & Fuller, 1979), but this result is diagnostic only — the fitted model's order is never adjusted based on it, and d remains 0 regardless of the test outcome.

| Constant | Value | Rationale |
|---|---|---|
| Order (p, d, q) | Fixed at (1, 0, 1) | Not automatically selected; researchers should verify this specification suits their series rather than assuming an optimal order was searched (contrast with automatic approaches, Hyndman & Khandakar, 2008). |
| enforce_stationarity, enforce_invertibility | Both False (relaxed from SARIMAX defaults) | Gives the optimizer more numerical freedom to converge on difficult series, at the cost of no longer guaranteeing a stationary/invertible fitted representation (Hamilton, 1994). |
| Cross-validation | Walk-forward (TimeSeriesSplit), folds fit in parallel via joblib | Same leakage-prevention rationale as the ML time-series modules (Bergmeir & Benítez, 2012); folds are independent, so parallelizing them is a pure speed optimization. |
| Minimum sample guard | max(10, cv_folds * 5) | More lenient than the ML modules' guard; very short series will still produce unstable order estimates even if accepted. |

Because the model never differences the series, fitting on data the ADF test flags as non-stationary risks the classical spurious-regression problem (Granger & Newbold, 1974). Researchers are strongly encouraged to check the reported adfuller_p/stationary fields and, if the series is flagged non-stationary, difference it themselves before upload — the platform will not do this automatically. Reported coefficients (ar.L1, ma.L1, and any exogenous-variable terms, plus sigma2, the estimated innovation variance) do not have the same "predictor effect" interpretation as an OLS coefficient.

### *11.4.2 LASSO Regression*

See **Section 10.4.4** for the general methodology and citations behind LASSO; settings specific to this data structure follow.

| Constant | Value | Rationale |
|---|---|---|
| Regularization strength (alpha) | Selected automatically via cross-validation over 50 candidate values | Data-driven selection rather than a fixed penalty; the selected value is reported as alpha_selected. |
| Feature scaling | StandardScaler, fit on training data only | L1 penalty is scale-dependent (Hastie et al., 2009). |
| Inner/outer CV | TimeSeriesSplit (walk-forward), no shuffling | Prevents lagged features built from future rows from leaking into training (Bergmeir & Benítez, 2012). |

| Constant | Value | Rationale |
| --- | --- | --- |
| Convergence | max_iter = 5,000 | Coordinate descent is iterative; guards against unconverged fits. |
| Random state | 42 | Reproducibility of the solver's coefficient path. |

### *11.4.3 Ridge Regression*

See **Section 10.4.5** for the general methodology and citations behind Ridge regression; settings specific to this data structure follow.

| Constant | Value | Rationale |
| --- | --- | --- |
| Alpha grid | 25 log-spaced values from 0.01 to 1,000, selected via cross-validation | Ridge has no natural maximum useful penalty the way LASSO does, so a wide manual grid is used (Hoerl & Kennard, 1970). |
| Feature scaling | StandardScaler, fit on training data only | Same scale-dependence rationale as LASSO. |
| Solver | Closed-form (sklearn default) | Ridge has an exact analytical solution; no iterative convergence risk, unlike LASSO. |
| Cross-validation | TimeSeriesSplit (walk-forward) | Same leakage rationale as LASSO (Bergmeir & Benítez, 2012). |

### *11.4.4 Bagging*

See **Section 10.4.6** for the general methodology and citations behind Bagging; settings specific to this data structure follow.

| Constant | Value | Rationale |
| --- | --- | --- |
| Base tree regularization | max_depth = 6, min_samples_leaf = 5 | An unconstrained base tree overfits noise in a small feature set (Hastie et al., 2009, Ch. 8). |
| max_samples / max_features | 0.8 / 0.8 | Row and column subsampling decorrelates trees; without it, bootstrap resampling alone gives limited diversity (Breiman, 1996). |
| n_estimators | 200 | Reduces variance of the averaged prediction; more trees give diminishing returns. |
| Feature importance | Permutation importance (n_repeats = 10) | BaggingRegressor has no native impurity-based importance; permutation importance also avoids the high-cardinality bias of impurity |

| Constant | Value | Rationale |
|---|---|---|
| | | measures (Strobl, Boulesteix, Zeileis, & Hothorn, 2007). |

### *11.4.5 Random Forest*

See **Section 10.4.7** for the general methodology and citations behind Random Forest; settings specific to this data structure follow.

| Constant | Value | Rationale |
|---|---|---|
| max_features | "sqrt" — square root of the number of predictors | The defining Random Forest mechanism; randomizes candidate predictors at every split (Breiman, 2001). |
| max_depth, min_samples_leaf, min_samples_split | 8, 5, 10 | Bounds tree growth; an unconstrained tree with a small lag-feature set memorizes noise rather than a stable pattern. |
| n_estimators | 200 | Same variance-reduction rationale as Bagging. |
| Feature importance | Native impurity-based importance | Standard sklearn output; carries a known bias toward high-cardinality/correlated predictors (Strobl et al., 2007). |

### *11.4.6 Gradient Boosting*

See **Section 10.4.8** for the general methodology and citations behind Gradient Boosting; settings specific to this data structure follow.

| Constant | Value | Rationale |
|---|---|---|
| learning_rate | 0.05 | Shrinks each tree's contribution; a lower rate trades more required trees for better generalization (Friedman, 2001). |
| max_depth | 3 | Standard shallow-tree "weak learner" depth for boosting. |
| subsample | 0.8 | Stochastic gradient boosting — each tree trains on a random subsample of rows, adding regularization (Friedman, 2002). |
| Early stopping | validation_fraction = 0.1, n_iter_no_change = 10, tol = 1e-4 | Halts boosting once held-out performance stops improving, preventing the model from |

| Constant | Value | Rationale |
| --- | --- | --- |
| | | continuing to fit noise (Zhang & Yu, 2005). |
| n_estimators (max / used) | 300 requested; actual count reported separately | Early stopping typically uses fewer than the requested maximum; the count actually used is the figure that reflects true model complexity. |

### 11.5 Reading Your Results

Coefficient, standard-error, and feature-importance fields follow the same format described in Section 10.5. Time series output additionally includes:

- Forecasted values for the target variable.
- Cross-validation metrics reported as an average (mean) and a spread (standard deviation) across folds — this tells you how stable the model's out-of-sample accuracy is. A small spread means dependable performance.

## 12. Panel Data Analysis

Use this module when you have many subjects observed across multiple time periods — for example, 40 firms tracked annually for 15 years.

### 12.1 Available Methods

| Method | Best used for |
| --- | --- |
| Fixed Effects | Controls for stable, unobserved traits of each subject that don't change over time. |
| Random Effects | Assumes subject-level differences are uncorrelated with your predictors. |
| LASSO / RIDGE | Regularized regression with many predictors. |
| FOREST / BOOSTING / BAGGING | Tree-based ensemble models. |

This table summarizes when to use each method. For the underlying methodology, constants, and citations behind every method, see **Section 12.4** below.

### 12.2 Screen Elements

Panel analysis combines cross-sectional and time controls:

| Element | What it does |
| --- | --- |
| Upload File | Select your .csv. |
| Methods | Choose the estimation method. |

| Element | What it does |
|---|---|
| Id of the Data | The column identifying each subject (firm/person/region). Required to separate subjects. |
| Dependent Variable | The outcome. |
| Independent Variables | Predictors (multi-select). |
| Categorical Variables | Category predictors, encoded automatically. |
| Outliers | Detect/treat extreme values, or leave as-is. |
| Date Variable | The time-period column (year, quarter, month). This is what makes the data a panel. |
| Summary Statistics / Predict / Clear | As in the other modules. |
| Download | Exports results to PowerPoint. |

## 12.3 Procedure — Running a Panel Model

33. Open the Panel Data module.
34. Upload your CSV.
35. Choose a Method (e.g. Fixed Effects).
36. Set Id of the Data to your subject identifier.
37. Set the Dependent Variable.
38. Add Independent Variables (and Categorical Variables if any).
39. Choose an Outliers option.
40. Select the Date Variable (the time column). If it isn't valid dates, a warning appears.
41. (Recommended) Click Summary Statistics.
42. Click Predict.
43. Review results and diagnostics; Download if desired.
44. Use Clear to reset.

**Note**

The key difference from cross-sectional analysis is the Date Variable: the same subject ID appears on multiple dates. Make sure your ID and date columns are both correct, or the panel structure won't be recognized.

## 12.4 Estimation Methods: Constants and References

This section documents, for every method listed in Section 12.1, the specific constants (hyperparameters), validation design, diagnostic tests, and methodological literature the implementation draws on.

### *12.4.1 Fixed Effects*

Fixed Effects is fit as OLS on within-transformed (entity-demeaned) data, following the standard panel-data approach for controlling for stable, unobserved entity characteristics (Wooldridge, 2010; Baltagi, 2021). Cluster-robust standard errors, clustered by entity, correct for the within-entity correlation that ordinary standard errors would ignore (Cameron & Miller, 2015).

| Constant | Value | Rationale |
|---|---|---|
| Degrees-of-freedom correction | n_observations - n_entities - k | The within transformation implicitly estimates one fixed effect per entity even though they are not explicit regressors; ordinary OLS degrees of freedom would understate the true parameter count (Wooldridge, 2010). |
| Cross-validation | Time-based walk-forward (preferred) or entity-based GroupKFold (fallback) | FE cannot estimate an individual fixed effect for an entity it never trained on, so entity-based CV is an explicitly weaker approximation (average fixed effect substituted); time-based CV over known entities is the more meaningful generalization test. |
| Time-invariant variables | Automatically dropped (zero variance after demeaning), reported explicitly | A regressor that never changes within an entity provides no identifying information to a within estimator (Wooldridge, 2010). |

Diagnostics reuse the Breusch-Pagan/White heteroskedasticity tests, VIF/correlation multicollinearity checks, and Jarque-Bera normality test from the cross-sectional modules, plus two panel-specific tests not applicable outside a panel setting: a serial-correlation test that regresses each entity's residual on its own lag (never crossing entity boundaries), in the spirit of panel-specific serial correlation testing (Wooldridge, 2010), and a cross-sectional dependence test using Pesaran's CD statistic, computed from the average pairwise correlation between entities' residual series (Pesaran, 2004). Where cross-sectional dependence is detected, Driscoll-Kraay standard errors are the standard recommended remedy (Driscoll & Kraay, 1998). Durbin-Watson, RESET, and Cook's Distance from the OLS module are intentionally not carried over — the panel-specific tests are the structurally appropriate substitutes, not omissions.

### *12.4.2 Random Effects*

Random Effects treats entity-specific effects as random draws, uncorrelated with the regressors, and is estimated via Feasible GLS using quasi-demeaning: rather than subtracting each entity's full group mean (as Fixed Effects does), a per-entity weight theta — between 0 and 1 — determines how much of the entity mean is subtracted (Wooldridge, 2010, Ch. 10). The two variance components theta depends on (within-entity variance sigma-squared-e, and between-entity variance sigma-squared-u) are estimated using the Swamy-Arora method (Swamy & Arora, 1972), and each entity's random effect is recovered as a Best Linear Unbiased Predictor (BLUP) that shrinks its average residual toward zero, with less shrinkage for entities observed over more time periods (Wooldridge, 2010).

| Constant | Value | Rationale |
|---|---|---|
| theta (quasi-demeaning weight) | 1 - sqrt(sigma2_e / (T_i * sigma2_u + sigma2_e)), per entity | Interpolates between pooled OLS (theta near 0) and full Fixed Effects demeaning (theta near 1) depending on how much random-effect variance is present relative to noise, and how many periods that entity was observed. |
| rho | sigma2_u / (sigma2_u + sigma2_e) | The share of total residual variance attributable to between-entity differences. |
| Unseen-entity prediction | BLUP = 0 | The theoretically correct RE prediction, not an approximation: since entity effects are modeled as mean-zero random draws, the expected effect for a never-observed entity is exactly zero. As a result, time-based and entity-based cross-validation are equally valid for RE, unlike for Fixed Effects. |
| Hausman test | Compares FE and RE coefficient vectors; $p < 0.05$ rejects RE in favor of FE | The standard test for choosing between FE and RE (Hausman, 1978); computed here using the Moore-Penrose pseudoinverse of the covariance difference matrix, a numerical safeguard since that matrix is not guaranteed to be well-conditioned in finite samples (Baltagi, 2021). |

Diagnostics are identical in format to the Fixed Effects module (heteroskedasticity, multicollinearity, normality, serial correlation, cross-sectional dependence), applied to the quasi-demeaned residuals rather than the fully demeaned ones.

See **Section 10.4.4** for the general methodology and citations behind LASSO; settings specific to this data structure follow.

### *12.4.3 LASSO Regression*

Panel LASSO applies the within (fixed-effects) transformation described in Section 9.1 before fitting, so that regularization operates on within-entity variation only — this is the panel-data analogue of Fixed Effects estimation combined with variable selection (Wooldridge, 2010). Because LASSO performs genuine variable selection, EconWebCast follows standard post-selection practice (Belloni & Chernozhukov, 2013): an ordinary least squares model is refit using only the variables LASSO retained, and cluster-robust standard errors (clustered by entity) are computed from that refit (Cameron & Miller, 2015). These cluster-robust values are reported separately from the raw LASSO coefficients, which — consistent with every other LASSO output in this platform — carry no valid classical standard errors of their own (Lee, Sun, Sun, & Taylor, 2016).

| Constant | Value | Rationale |
| --- | --- | --- |
| Alpha | User-specified, default 1.0 | Not automatically cross-validated for selection in the panel module; cross-validation reports the stability of the chosen value rather than searching for an optimum. |
| Cross-validation method | Time-based walk-forward (preferred) or entity-based GroupKFold (fallback) | Selected automatically based on whether a usable date column is available; see Section 9.2. |
| Demeaning scope | Computed from each fold's training rows only | Prevents test-row information from leaking into how training rows are transformed. |
| Post-selection inference | OLS on retained variables, cluster-robust SEs by entity | Standard post-LASSO practice (Belloni & Chernozhukov, 2013); accounts for within-entity correlation (Cameron & Miller, 2015). |

See **Section 10.4.5** for the general methodology and citations behind Ridge regression; settings specific to this data structure follow.

### *12.4.4 Ridge Regression*

Panel Ridge uses the same within-transformation and dual cross-validation design as Panel LASSO (Section 12.4.3). Because Ridge never sets coefficients to zero, there is no natural post-selection subset to refit — the platform's cluster-robust standard errors are computed from an unpenalized OLS fit on the full set of within-transformed variables, clustered by entity (Cameron & Miller, 2015). Researchers should note that these standard errors describe an unpenalized estimator and are not formally the sampling variability of the Ridge coefficients themselves, which require dedicated ridge-inference methods for fully rigorous testing (Obenchain, 1977; Cule & De Iorio, 2013). The reported coefficients remain the correct, shrinkage-adjusted point estimates; the accompanying p-values should be read as an approximate reference derived from the corresponding unpenalized model, not a precise test of the Ridge estimate.

| Constant | Value | Rationale |
| --- | --- | --- |
| Alpha | User-specified, default 1.0 | Same as Panel LASSO — not automatically selected via cross-validation. |
| Retained variables | All (Ridge never zeros a coefficient) | Defining behavioral difference from Panel LASSO, useful for direct comparison of the two outputs. |
| Entity fixed effects | alpha_i = mean(y_i) − mean(X_i)·beta | The correct fixed-effect definition, accounting for the regressors, not simply the raw entity mean of y (Wooldridge, 2010). |

See **Section 10.4.6** for the general methodology and citations behind Bagging; settings specific to this data structure follow.

#### 12.4.5 Bagging

Consistent with the general panel design described in Section 9.1, Panel Bagging encodes entity identity as one-hot dummy features rather than demeaning the data, since a tree ensemble can split directly on entity identity to learn entity-specific baselines without the omitted-variable bias that makes demeaning necessary for linear models (Athey & Imbens, 2019). Individual entity-dummy importances are summed into a single reported entity_id importance figure, analogous to asking how much of the model's explanatory power comes from fixed effects versus substantive predictors in a linear specification. An additional ensemble-stability diagnostic reports the standard deviation and mean bias of individual base-tree predictions around the ensemble average — a measure of tree agreement unique to bagged ensembles.

See **Section 10.4.7** for the general methodology and citations behind Random Forest; settings specific to this data structure follow.

#### 12.4.6 Random Forest

Panel Random Forest follows the same entity-dummy, no-demeaning design as Panel Bagging (Section 12.4.5) and explicitly sets max_features = "sqrt", correctly implementing per-split feature randomization for panel data (Breiman, 2001). Three related but distinct fit-quality figures are reported together: in-sample $R^2$ (fit on the training data itself, typically the most optimistic), out-of-bag $R^2$ (a near-free, bootstrap-based estimate of out-of-sample performance), and cross-validated $R^2$ (from fully independent, re-partitioned train/test splits, the most rigorous of the three). Comparing all three — rather than relying on any single figure — gives a fuller picture of whether the model is overfitting (Kohavi, 1995).

See **Section 10.4.8** for the general methodology and citations behind Gradient Boosting; settings specific to this data structure follow.

#### 12.4.7 Gradient Boosting

Panel Boosting follows the same entity-dummy design as Panel Bagging and Panel Forest. In addition to the standard outputs, this module reports the complete training deviance curve (loss at every boosting iteration) and a staged $R^2$ trajectory sampled at up to twenty checkpoints along the boosting sequence, computed on the full training sample (Friedman, 2001). Both are useful for visually confirming smooth convergence, but — because they are computed in-sample — they should not be read as evidence of good generalization on their own; the reported cross_validation block remains the honest out-of-sample estimate.

### 12.5 Reading Your Results

Coefficient and feature-importance fields follow the same format described in Section 10.5. Panel models additionally report:

- Entity fixed effects or entity random effects — a per-entity intercept. Fixed Effects estimates this directly from each entity's data; Random Effects produces a shrunk, empirical-Bayes estimate (a Best Linear Unbiased Predictor, or BLUP). See Section 12.4.1 and Section 12.4.2 for the exact definitions.
- Cluster-robust standard errors, for methods that report them (Fixed Effects, Random Effects, LASSO, Ridge) — standard errors adjusted for the fact that repeated observations of the same entity are correlated.
- Panel-specific diagnostics — serial correlation and cross-sectional dependence — described alongside Fixed Effects in Section 12.4.1. As with the other modules, every test ends in a plain-language conclusion.

## 13. Exporting Results to PowerPoint

Every analysis module can export results to a formatted PowerPoint deck, ready to drop into a paper, thesis, or presentation.

### Procedure — Exporting

45. Run a successful Predict (or Summary Statistics).
46. Locate the download button that appears above/near the results table (labeled Download as PowerPoint or a download icon).
47. Click it. Your browser downloads a .pptx file.
48. Open it in PowerPoint, Google Slides, or Keynote.

What the deck contains: your model summary, coefficients or feature importance, fit statistics, and diagnostic tests (for model runs), or the descriptive table (for summary statistics).

> **Tip**
>
> Run Summary Statistics and Predict separately if you want both a descriptive-statistics slide and a results slide to export.

## 14. Worked Examples

The following realistic scenarios illustrate end-to-end use. Substitute your own data and columns.

### Example A — Returns to education (Cross-Sectional, OLS)

**Research question:** *Does an additional year of schooling raise hourly wages?*

Data: One row per worker, with columns worker_id, wage, education_years, experience, region.

49. Open Cross-Sectional Data → Upload the CSV.
50. Method: OLS. ID Column: worker_id. Dependent Variable: wage.
51. Independent Variables: education_years, experience. Categorical Variables: region.
52. Outliers: Yes. Click Summary Statistics to confirm wages look sensible.
53. Click Predict.

Expected result: A positive, statistically significant coefficient on education_years would indicate that more schooling is associated with higher wages, holding experience and region constant. Check the Breusch–Pagan test; if heteroskedasticity is flagged, note that robust standard errors are advisable.

### Example B — Forecasting monthly rental prices (Time Series, ARIMA)

**Research question:** *What is the expected trajectory of average rents?*

Data: One row per month, with month (dates) and avg_rent.

54. Open Time Series Data → Upload.
55. Method: ARIMA. Date Variable: month. Confirm the green validation (no warning).
56. Leave Start/End Date at the full range. Endogenous Variable: avg_rent.
57. Click Line Graph to eyeball the trend, then Predict.

Expected result: Forecasted rent values plus cross-validation metrics. A low standard deviation across folds means the forecast is stable; a high one means treat the forecast with caution.

### Example C — Effect of a policy across countries (Panel, Fixed Effects)

**Research question:** *Did a tax reform change investment, controlling for fixed country traits?*

Data: One row per country-year, with country_id, year, investment, tax_rate, gdp_growth.

58. Open Panel Data → Upload.
59. Method: Fixed Effects. Id of the Data: country_id. Date Variable: year.
60. Dependent Variable: investment. Independent Variables: tax_rate, gdp_growth.
61. Outliers: No. Summary Statistics, then Predict.

Expected result: The coefficient on tax_rate estimates its effect on investment within countries over time, netting out stable country differences.

## 15. Common Errors and Troubleshooting

| Symptom | Likely cause | What to do |
|---|---|---|
| Predict button is greyed out | A required field is missing. | Ensure the method, ID/date column, dependent variable, and at least one independent variable (and outlier choice, where shown) are all selected. |
| "Selected column does not contain valid date values" | The chosen Date Variable has non-date entries. | Pick the correct date column, or reformat the dates (e.g. 2023-01-31) and re-upload. |
| "Something went wrong. Try again!!" | The model could not be estimated with the current inputs. | Check for too many missing values, non-numeric data in numeric columns, or too few rows. Clean the data and retry. |
| "Not enough observations..." | The dataset is too short for the requested validation. | Provide more rows, or reduce the number of cross-validation folds if available. |
| "Too many requests. Please try again in one minute." | You ran many analyses in quick succession (rate limit). | Wait about a minute, then continue. Space out large batches of runs. |
| Asked to sign in again / "Missing or invalid authorization" | Your session expired. | Sign in again with Google and re-run. |
| Dropdowns are empty after upload | The file isn't a valid CSV or lacks a header row. | Re-export as CSV with column names in the first row. |
| Numbers look wrong in Summary Statistics | Numeric columns contain symbols or text. | Use Data Cleanup → Convert Text to Numeric, or clean the source file. |

| Symptom | Likely cause | What to do |
|---|---|---|
| Empty or flat Line Graph | Wrong date/target column, or missing values. | Re-check the Date Variable and Endogenous Variable selections. |

**Note**

If an error persists after checking the table above, click Clear, re-upload a small, clean subset of your data, and confirm it runs. Then add complexity back gradually to isolate the problem.

## 16. Best Practices for Researchers

Follow these recommendations to get results you can defend.

### Before you model

62. Keep a pristine original. Never overwrite your raw data file; clean on copies.
63. Name columns clearly. You'll select them by name — clarity prevents mistakes.
64. Inspect first. Always run Summary Statistics (and Line Graph for time series) before Predict.

### Choosing a method

65. Match the model to the data structure. Cross-sectional, time series, and panel are not interchangeable — use the right module.
66. Match the model to the outcome. Use Logit for yes/no outcomes; OLS/GLS for continuous outcomes.
67. Compare models. Run OLS, then LASSO/Ridge, then a tree-based model on the same data and compare fit and stability before concluding.

### Interpreting results

68. Read the diagnostics, not just the coefficients. If heteroskedasticity or multicollinearity is flagged, adjust your interpretation accordingly.
69. Distinguish significance from importance. A significant coefficient can still be small; look at the magnitude, not only the p-value.
70. Prefer stable forecasts. In time series, trust models whose cross-validation spread is small.

### Reporting

71. Export as you go. Download the PowerPoint immediately after a good run so your results and diagnostics stay together.
72. Record your settings. Note the method, variables, date range, and outlier choice for each result so it's reproducible.
73. Report diagnostics alongside coefficients. Reviewers of empirical work expect to see that assumptions were checked, not just the headline results.

## 17. Quick Reference

### Module selector

| Your data | Module | Common first method |
|---|---|---|
| Many subjects, one time point | Cross-Sectional Data | OLS |
| One subject, many time points | Time Series Data | ARIMA |
| Many subjects, many time points | Panel Data | Fixed Effects |

## 18. Contact, Feedback, and Ongoing Development

EconWebCast is under continuous development. New estimation methods, diagnostics, and usability improvements are added over time based on real researcher feedback.

If you have questions about how to use a feature, encounter an issue or unexpected error, or have suggestions for models, diagnostics, or workflow improvements you'd like to see added, please reach out.

**Email:** kbretmanna@gmail.com

**Application:** https://econ-web-cast.vercel.app/

Comments, bug reports, and feature requests from the research community are highly encouraged and genuinely help guide the roadmap.

## References

In-text citations throughout this guide follow the author–date format common to applied econometrics and statistics publications (e.g., Author, Year), so that the methodological claims behind each model's default settings can be traced to their source. The full reference list below is formatted in APA style (7th edition conventions) and covers the statistical and machine-learning methods, diagnostic tests, and validation techniques implemented across EconWebCast's Cross-Sectional, Time Series, and Panel Data modules. Researchers preparing a manuscript that reports EconWebCast output are encouraged to cite the specific method(s) used from this list, in addition to citing EconWebCast itself and noting the application version (see the cover page).

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**A closing note**

This guide describes EconWebCast as of Version 1.0. The application continues to evolve — screens, methods, and outputs may be refined in future releases. For the current version, always refer to the in-app "Guide Me" help and feel free to write in with questions or feedback at kbretmanna@gmail.com.

*End of User Guide*